

Roktim Mascharak

PERSONAL DATA

PhD. Student, London School of Geometry and Number Theory(UCL,KCL,ICL). Email: roktim.mascharak.23@ucl.ac.uk (Alternate emails: r.mascharak23@imperial.ac.uk/roktim.mascharak@kcl.ac.uk)

AREA OF INTEREST

Algebraic Geometry, Birational Geometry, especially the Minimal Model Program and Birational Geometry of Foliation.

EDUCATION

October 2023-Present	PhD in Mathematics at London School of Geometry and Number Theory (UCL,KCL,ICL).
August 2020-July 2023	MSc. in Mathematics at Tata Institute of Fundamental Research Homi Bhabha Road , Colaba Mumbai ,India
August 2017-March 2020	B.Sc (Honours) in Mathematics at Presidency University , Kolkata, India Major: Mathematics

ACADEMIC ACHIEVEMENTS

October 2023-Present	Studentship at London School of Geometry and Number Theory(Joint PhD Program at University College London, King’s College London, and Imperial College London).
August 2020-2023 July 2020	Recipient of Institute fellowship at Tata Institute of Fundamental Research Mumbai. Secured All India Rank 128 in IIT-JAM Mathematics

SEMINARS, INTERNSHIPS AND TRAINING CAMPS ATTENDED

May 2026	Simon’s Collaboration in Moduli of higher dimensional Varieties, Ann Arbor
April 2026	Equivariant K-stability, Edinburgh
March 2025	Winter School in K-stability,CIRM, Marseille.
July 2025	Summer Research Institute in Algebraic Geometry 2025, Colorado(Bootcamp and main Conference).
August 2024	Birational Geometry and Foliations at Shanghai
September 2024-December 2024	LSGNT lunch seminar (Organizer)
May 2022	Advance Instructional School in Mathematics on Intersection Theory at IIT Bombay
December2018	Mini MTTS Training Camp Kolkata

TALKS AND POSTERS PRESENTED

June 2026	CALF Seminar, Nottingham:-Invitation to Foliated MMP for log canonical foliations
May 2026	Birational Geometry Seminar, UCLA(Online Talk):- Introduction to Foliated MMP for log canonical foliations
November 2025	Algebraic Geometry North Eastern Series, Amherst (Poster)
October 2025	Moduli Problems in Birational Geometry and Foliation Theory, Pisa (Poster)
September 2025	New Trends in Moduli Inspired by K-Stability, Nottingham:- Invitation to Foliated MMP.
July 2025	Summer Research Institute in Algebraic Geometry 2025, Colorado (Poster)
March 2025	Winter School in K-stability,CIRM(5 minutes talk)
November 2024	Junior Geometry Seminar at Imperial College London.
September 2024	QED. Seminar at Presidency University Kolkata.
June 2024	LSGNT first year talk on Sarkisov Program for Foliations at University College London
October 2022-December 2022	-MMP Seminar (As Part of Student Algebraic Geometry Seminar) at T.I.F.R. Mumbai
September 2022-January 2023	-BCHM (Existence of Minimal Model for Varieties of Log General Type) Learning Seminar at T.I.F.R. Mumbai (Co-Organizer with Dr.Priyankur Chaudhuri)
June 2022	-Existence Of Flips and Minimal Model Program (Master's Thesis Presentation) at T.I.F.R Mumbai
April 2021	-Given a talk on Thom Isomorphism at T.I.F.R Mumbai
November 2022	Inversion of Adjunction (As a part of Algebraic Geometry Pre-print Seminar) at T.I.F.R Mumbai. This talk is based on a recent work of O. Fujino and K. Hashizume, this is the arxiv link to the paper https://arxiv.org/abs/2107.04854 .

PRE-PRINTS AND PUBLICATIONS

1. *On Log Canonical MMP for Kähler 3-Fold* <https://arxiv.org/abs/2306.07539>
2. *ACC for lc thresholds for algebraically integrable foliations* Joint work with Omprokash Das and Jihao Liu. (Accepted for publication at Selecta Mathematica) <https://arxiv.org/abs/2307.07157>.
3. *On The Log Sarkisov Program For Foliations On Projective 3-Folds* (Published at Math. Z. 312, 124 (2026). <https://doi.org/10.1007/s00209-026-04001-y>)<https://arxiv.org/abs/2406.09434>
4. *Log Canonical Minimal Model Program for Co-rank One Foliations on Threefold.* <https://arxiv.org/abs/2410.05178> Joint work with Priyankur Chaudhuri. (Submitted)

PRE-PRINTS IN PREPARATION

- Log Sarkisov Program for Kähler 3-fold with klt singularities and applications. Joint work with Swapnajit Das.
- Crepant foliated modifications. Joint work with Paolo Cascini and Calum Spicer.

ON-GOING PROJECTS

- MMP for adjoint Foliated structures in dimension three and applications. Joint work with Stefania Vassiliadis.
- MMP for co-rank foliations on four-folds with non pseudoeffective $K_{\mathcal{F}}$. Joint work with Priyankur Chaudhuri.

TEACHING EXPERIENCE

January 2026- Present	Senior Teaching Assistant for Analysis 2 at Imperial College London
October 2025- December 2025	Teaching Assistant for Linear Algebra and Groups at Imperial College London
October 2025	Teaching assistant for Introduction to University Mathematics at Imperial College London.
January 2025- April 2025	Senior Teaching Assistant for Analysis 2 at Imperial College London
January 2025- April 2025	Teaching Assistant for Calculus 2 at King's College London
October 2024- December 2024	Teaching Assistant for Analysis 2 at Imperial College London
October 2024- December 2024	Teaching Assistant for Linear Algebra and Groups at Imperial College London
October 2024	Teaching assistant for Introduction to University Mathematics at Imperial College London.
January 2024- March 2024	Teaching Assistant for the Course Analysis 2 at Imperial College London.
May 2023	TA for the mini courses on Minimal Model Program and Rationality properties of Algebraic Varieties at Advance Instructional School on Birational Geometry at Chennai Mathematical Institute.

LANGUAGE

Reading and Writing Proficiency Bengali (Mother Tongue), Hindi, English

EXTRACURRICULAR ACTIVITIES

Participated in TIFR Frontier of Science Outreach Program.

Taught Mathematics to school students at Mumbai Math Circle (Organized by T.I.F.R Mumbai)